

⚠ FACILITATOR COPY — CONTAINS ANSWER KEYS & GRADING GUIDANCE

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This document is for the OAO 3rd Grade Math teacher/facilitator only.



OPTIMA ACADEMY ONLINE

3rd Grade Math Teacher Manual

Facilitator Guide · Singapore Math / Florida B.E.S.T. Aligned

□ 12 Units · 75 Lessons

□ 12 Concept Checks

□ 12 Summative Assessments

□ 16 Spotlight Tasks (Weeks 17–32)

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How to Use This Manual

This manual documents the OAO 3rd Grade Math course as it is built and deployed on GitHub Pages, embedded in Canvas LMS as iframes. Every lesson, assessment type, and grading note in this document is drawn from the actual built course — not from assumption.

The manual is organized in reading order below. Most facilitators will use it in three ways: **before the course starts** (read Sections 1–3 to understand what students experience), **weekly** (Section 7 for grading workflow and announcement templates), and **at grading time** (Sections 5 and 6 for answer guides and rubrics).

- ☐ **Teacher-only content.** Sections 5 and 6 contain answer keys for Concept Checks and Summative Assessments. Do not post these pages in Canvas or share them with students or families. Keep this document in your facilitator folder only.

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How This Course Works

Lesson structure · typical week · major assessments

The Course at a Glance

OAO 3rd Grade Math follows Dimensions Math (Singapore Math) scope and sequence, aligned to Florida B.E.S.T. standards. The course runs 12 units across the school year. Every lesson is a self-contained HTML page hosted on GitHub Pages and embedded

in Canvas as an iframe. Students work entirely independently — there is no synchronous class session. The teacher's role is to assign work in Canvas, grade paper-based assessments, and support students who need help.

Component	Count	Where it lives	Who grades it
Lesson pages (HTML)	75 lessons across 12 units	GitHub Pages → Canvas iframe	Self-graded / ungraded
Concept Checks (CC)	1 per unit (12 total)	PDF → printed or Canvas upload	Teacher
Summative Assessments (SA)	1 per unit (12 total)	PDF → printed or Canvas upload	Teacher
Independent Practice (IP)	1 per lesson (~75 total)	PDF → printed or Canvas upload	Teacher (spot-check) or completion
Spotlight Tasks	2 per week, Weeks 17–32 (32 total)	HTML → Canvas iframe + upload	Teacher (VR capture/submission)

A Typical Lesson

Each lesson page (e.g., Lesson 3.01.01 "Place Value, Part 1") opens in the Canvas iframe. The student sees:

1. A **progress bar** at the top showing how far through the page they are.
2. A **header** with the lesson title, unit, lesson number, Florida B.E.S.T. standard, and focus skill.
3. An **Explorer Strip** ("Today's Mission") and **Essential Question + Objectives** card pair.
4. A **"You'll need" strip** (typically: Math Journal, Pencil, Ruler or straight edge).
5. Four **tabs**: Warm-Up → Learn → Practice → Show What You Know. (Explained in Section 2.)

The student moves through all four tabs in order. Tab checkmarks appear as they visit each tab. The Show What You Know tab has a turn-in checklist with confetti when all items are checked.

A Typical Week

What the student does	Page type	Graded?
Work through 1–2 lesson HTML pages	Lesson (iframe)	No — self-check only
Complete 1–2 Independent Practice PDFs	IP PDF (paper/upload)	Yes — completion or spot-check
During weeks 17–32: complete a Spotlight Task	Spotlight HTML + VR upload	Yes — teacher reviews submission
At end of unit: Concept Check	CC PDF (paper/upload)	Yes — teacher grades
At end of unit: Summative Assessment	SA PDF (paper/upload)	Yes — teacher grades

- ☐ **Pacing note.** The course does not prescribe an exact daily schedule. Most units have 6–10 lessons. A reasonable pace for a full-year course is 1–2 lessons per week, with a Concept Check at the unit midpoint and a Summative Assessment at unit end. Teachers set their own Canvas due dates.

Major Assessments by Unit

Unit	Title	Lessons	Concept Check (CC)	Summative Assessment (SA)
3.01	Numbers to 10,000	8	3.01.CC (after L1–L4)	3.01.SA
3.02	Mental Addition & Subtraction	10	3.02.CC	3.02.SA
3.03	Addition & Subtraction with Regrouping	6	3.03.CC	3.03.SA

3.04	Multiplication & Division (Review)	9	3.04.CC	3.04.SA
3.05	Multiplication (3-Digit $\times$ 1-Digit)	7	3.05.CC	3.05.SA
3.06	Division (2–3-Digit $\div$ 1-Digit)	7	3.06.CC	3.06.SA
3.07	Data — Graphs & Tables	2	3.07.CC	3.07.SA
3.08	Multiplication Tables 6–9	7	3.08.CC	3.08.SA
3.09	Fractions of a Whole & Number Line	4	3.09.CC	3.09.SA
3.10	Equivalent Fractions & Operations	7	3.10.CC	3.10.SA
3.11	Measurement (Length, Capacity, Mass)	7	3.11.CC	3.11.SA
3.12	Geometry (Circles, Angles, Shapes)	7	3.12.CC	3.12.SA

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The Lesson Page Tour

Four tabs — what students do in each

Every lesson follows the same four-tab structure. The tab bar sticks at the top as the student scrolls. A checkmark (✓) appears on each tab after the student visits it. The tabs are designed to guide students through a complete Singapore Math lesson arc: readiness → concept → practice → application.

☐ Warm-Up

☐ Learn

☐ Practice

☐ Show What You Know

← Students move left to right through these four tabs. Each tab's checkmark lights up when visited.

□ Tab 1 — Warm-Up

What's here: A timed or untimed fluency drill (addition, subtraction, or multiplication facts), a Readiness Check (a quick concrete/pictorial problem at the prior grade level), and a Think Prompt with a Notice/Wonder grid and a prediction box.

Purpose: The Readiness Check connects to what students already know (the bridge from 2nd grade or the prior lesson). The Notice/Wonder prompt activates curiosity before the concept is taught — students guess and record a prediction. This is the classical narration-before-instruction move: observe first, then learn.

Teacher note: None of this tab is collected or graded. Journal entries auto-save in the browser but are not submitted anywhere. If a student skips this tab, it shows up indirectly in their performance on the Practice tab — don't chase it as a missing assignment.

□ Tab 2 — Learn

What's here: A concept video placeholder (placeholder text indicates video slots — videos are added as they are recorded), vocabulary flip cards (tap to reveal definition), a Key Concept box, interactive models (place-value mats, digit tiles, fraction strips, etc.), a Worked Example narrated by the Math Crew characters, a Socratic question, and a journal reflection prompt.

The CPA arc: Every Learn tab moves through Concrete (Touch It) → Pictorial (See It) → Abstract (Write It). Students build with on-screen manipulatives, read a chart representation, then write the standard/expanded form. This mirrors the Singapore Math CPA sequence.

Teacher note: The concept videos show as placeholders in lessons where they haven't been recorded yet. Students who encounter a video placeholder should still work through the rest of the tab — all the key concept content is also in the interactive and text elements on the page.

□ Tab 3 — Practice

What's here: Typically three interactive activities, each with its own Check button. When a student answers correctly, a green "pass" branch appears with positive feedback and the next-step nudge. When they answer incorrectly, an orange "retry"

branch appears that re-teaches the specific concept and resets the problem for another attempt.

Teacher note: These are fully self-graded. Students can retry indefinitely. No attempt data is sent to Canvas — the practice is formative, not summative. A student who can pass all three activities in Practice is generally ready for the Concept Check on that lesson's skill.

Tab 4 — Show What You Know

What's here: A short set of written/drawn problems (usually 2–4), a "Connection Forward" note linking today's skill to the next lesson, a copy-your-work or PDF-download option, and a turn-in checklist. When all checklist items are ticked, confetti fires.

Teacher note: The Show What You Know tab is the bridge to the Independent Practice PDF. Students who use it well will copy their journal work and compare it to the IP problems. The tab itself is not collected — the Independent Practice PDF (submitted separately in Canvas) is the graded artifact.

-  **Biggest misconception for new teachers:** The lesson page does not produce a submission. Students work on the HTML page, then download/print the Independent Practice PDF and submit that separately. If Canvas shows "no submission" for a lesson, check whether the IP PDF was submitted — not whether the iframe was opened.

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The Math Crew

Four characters who model mathematical thinking

Every lesson's Learn tab and Worked Example sections feature four student characters called the Math Crew. They appear in thought bubbles (side-by-side panels) whenever a new problem is worked through. Their purpose is to model the range of valid strategies and show that mathematics has multiple entry points.

A — Atlas (Purple)

The Builder

Uses place-value mats, discs, and concrete tools. "I like to start with pieces I can see and touch." Represents the Concrete phase of CPA.

H — Hazel (Green)

The Diagrammer

Draws number lines, bar models, and charts. "I draw it out — a diagram helps me see." Represents the Pictorial phase of CPA.

F — Felix (Gold)

The Shortcutter

Looks for mental math strategies and patterns. "I find the quickest way — sometimes I do it in my head." Bridges Pictorial to Abstract.

C — Cora (Blue)

The Checker

Verifies answers and catches errors. "I make sure we're right — does the answer actually check out?" Represents mathematical precision and self-monitoring.

- **Why it matters for the teacher:** When a student consistently uses only one strategy on their IP or assessments, you can nudge them toward another Crew member's approach. "Have you tried Hazel's way — can you draw a bar model for this?" The characters give you a shared vocabulary with students for talking about strategy flexibility without judgment.

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Ungraded Interactives

What the teacher does not collect or grade

The following elements live on the lesson HTML pages, give immediate feedback to students, and produce no Canvas submission. They are learning tools, not assessment items.

Interactive	Where it appears	What it does	Teacher action
Fluency drill	Warm-Up tab	10 fact-fluency problems, student types answers, gets instant right/wrong feedback	None — completion not tracked

Readiness Check	Warm-Up tab	1 prior-knowledge problem with reveal button	None
Notice/Wonder journal	Warm-Up tab	Free-text textarea, auto-saves in browser session	None — not submitted
Place-value mat	Learn tab	Drag/click to add/remove discs, number updates live	None
Vocabulary flip cards	Learn tab	Tap to flip front/back; term → definition	None
Digit tiles / match cards / value cards	Learn tab	Click-to-reveal or drag-to-match interactives	None
Socratic question + journal box	Learn tab	Text area for student thinking; auto-saves locally	None — not submitted
Practice activities (×3)	Practice tab	Self-check with pass/retry branches; infinite retries	None
Turn-in checklist + confetti	Show What You Know tab	Student self-check; celebration animation on completion	None

- ☐ **The key point:** A student who opens the lesson iframe and spends 30 minutes working through all four tabs has done real mathematical work — but nothing appears in the Canvas gradebook. The gradebook evidence comes from the Independent Practice PDF submission. These are complementary, not redundant. Don't penalize students for "missing" the iframe if the IP shows mastery, and don't award credit for opening the iframe if no IP was submitted.

Independent Practice PDFs (IP)

There is one IP PDF per lesson, named `math-lesson-3-XX-XX-independent-practice.pdf`. These are printed or uploaded student worksheets that parallel the lesson's Learn and Practice content. They are the primary formative assessment for each lesson.

Grading approach: IPs can be graded for accuracy (correct/incorrect per problem) or for completion (student showed work and attempted all problems). For most lessons, a spot-check of 3–4 key problems is sufficient. See the rubric in Section 6.

Concept Checks (CC)

One CC per unit, named `math-3-XX-CC-concept-check.pdf`. The CC is administered roughly at the unit midpoint (after the first 4 lessons of a longer unit, or after the full unit for shorter units). It is 4 pages, typically 8 problems, and covers the core concrete-to-abstract skills through models, standard form, expanded form, comparison, and explanation tasks.

Purpose: Catch gaps before the Summative Assessment. A student who misses 3+ problems on the CC should get targeted support before the SA.

Grading approach: Grade each problem right/wrong. For explanation questions (the final 1–2 items), use the rubric in Section 6. Total points and record in Canvas.

Summative Assessments (SA)

One SA per unit, named `math-3-XX-SA-summative-assessment.pdf`. The SA is 5 pages, 10 problems. It covers all unit skills and includes: show-your-work computation problems, a visual/drawing task, pattern/rule problems, word problems, an error-analysis item ("Find the Mistake"), and a multi-part reasoning item.

What makes the SA harder than the CC: The SA includes error analysis (identify and correct a wrong answer), multi-step reasoning, and "why are the answers different" explanation questions. These require full written responses, not just a number.

Grading approach: 1 point per problem for problems 1–8; 2 points for the final 2 reasoning/explanation problems. Maximum 12 points per SA. Use the answer guide in Section 6.

Spotlight Tasks (Weeks 17–32)

Spotlight Tasks are weekly VR activities embedded as iframes in Canvas. There are 32 Spotlight Task HTML files in the repository (two per week: a primary version and an alternate version labeled "-2"). The tasks run from Week 17 through Week 32 of the school year.



[VERIFY] Status at time of manual generation: The Spotlight Task HTML files currently contain placeholder content — the week-specific VR activity directions have not been added yet. The pages describe what the final VR activity will ask students to do (observe, build, model, or sort in VR, then upload a file). Once the final VR content is added to these pages, update this section with the specific task for each week. The grading workflow below applies to the final version.

Once VR content is live — grading approach: Each Spotlight Task asks students to upload a file from their VR work (a screenshot, export, or recording). Review the uploaded file for evidence that the student completed the activity. Grade on a 3-point rubric: 3 = file submitted and shows clear engagement with the math task; 2 = file submitted but connection to math is unclear; 1 = file submitted but activity appears incomplete; 0 = no submission.

The two versions per week ("-2" variant) are alternates — students do one version, not both. [VERIFY which version is primary for each week once VR content is finalized.]

Canvas Gradebook Setup

Assignment Group	Items	Grading	Suggested Weight
Independent Practice	~75 IP PDFs	Accuracy or completion; 10 pts each	30%
Concept Checks	12 CCs	Points per problem; 8 pts each	25%
Summative Assessments	12 SAs	12 pts each (see Section 6)	35%
Spotlight Tasks	16 weeks × 1 task	3-point rubric (see above)	10%

Weights are suggested. Adjust to match your school's gradebook policies. Mark IP as completion-scored if your school doesn't grade homework for points.

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Assessment Answer Guides & Rubrics

Teacher copy only — do not share with students

- ☐ This section contains answer keys for Unit 1 Concept Check and Summative Assessment, drawn from the built course. Answer keys for Units 2–12 should be generated when those assessments are finalized and added here.

General Grading Rubric — All Assessments

Use this rubric for any problem that asks a student to show work and explain. Apply it to the final 1–2 explanation problems on each CC and SA.

Score	Computation	Shown Work / Model	Explanation (when required)
Full credit	Correct answer	Work clearly shown; model matches answer	Uses math vocabulary; names the place or strategy; answer makes sense
Partial credit (½)	Minor error in computation but method is correct	Work partially shown; some steps visible	Explanation present but incomplete or missing vocab
No credit	Incorrect with no recoverable work	No work shown; answer only	No explanation or "I don't know"

IP Spot-Check Rubric (per lesson)

Score	Criteria
10 / Meets	All attempted; 80%+ correct on spot-check problems; work shown
7 / Approaching	Most attempted; 50–79% correct; some work shown
4 / Beginning	Half or fewer attempted; significant errors; little work shown

Unit 3.01 — Concept Check Answer Guide (3.01.CC)

Covers lessons 3.01.01–3.01.04: place value, standard/expanded form, comparing, number line

3.01.CC ANSWER KEY (8 PROBLEMS)

- 1.** For 3,521 → Thousands: **3**, Hundreds: **5**, Tens: **2**, Ones: **1**
- 2a.** 2,476 in expanded form: **2,000 + 400 + 70 + 6**
- 2b.** 4,000 + 300 + 50 + 8 in standard form: **4,358**
- 3.** 24 hundreds and 5 tens = 2,400 + 50 = **2,450** (key concept: 24 hundreds = 2 thousands 4 hundreds)
- 4a.** 3,486 < 3,498 (tens: 8 < 9)
- 4b.** 7,205 > 7,025 (hundreds: 2 > 0)
- 4c.** 4,800 = 4,800
- 5.** Order least to greatest: **4,025 < 4,205 < 4,520**
- 6.** 4,250 plotted at the halfway mark between 4,200 and 4,300 on a 4,000–5,000 number line with marks every 100. Accept any clearly labeled dot at 4,250.
- 7.** Draw place-value blocks for 1,235: 1 large square (thousand), 2 squares (hundreds), 3 lines (tens), 5 dots (ones). Accept labeled diagram. Answer: **1 thousand, 2 hundreds, 3 tens, 5 ones**
- 8.** Greater number: **2,809**. Explanation must say: same thousands digit (2), but hundreds digits differ — 8 hundreds vs. 0 hundreds, so 2,809 is greater. Award partial credit for correct answer without vocabulary.

Unit 3.01 — Summative Assessment Answer Guide (3.01.SA)

Covers all of Unit 1: place value, expanded form, comparing, number line, patterns, rounding ×3 (thousand, hundred, ten)

3.01.SA ANSWER KEY (10 PROBLEMS, 12 PTS TOTAL)

- 1.** 2,476 in expanded form: **2,000 + 400 + 70 + 6** (1 pt)
- 2.** 3 thousands, 4 hundreds, 7 tens, 8 ones → **3,478** (1 pt)
- 3a.** 5,367 < 5,376 (tens: 6 < 7) (1 pt for all three symbols)
- 3b.** 8,200 > 8,020 (hundreds: 2 > 0)

- 3c.** $3,499 < 3,500$
- 4.** Number line 4,000–5,000, marks every 100. 4,650 is between 4,600 and 4,700 (closer to 4,700). 4,200 is at the 4,200 mark. Accept clearly labeled dots. (1 pt)
- 5.** Pattern: 4,200 → 4,400 → 4,600 → **4,800** → **5,000**. Rule: +200 each time. (1 pt)
- 6.** Round 4,672 to nearest thousand: look at hundreds digit ($6 \geq 5$) → round up → **5,000** (1 pt)
- 7.** Round 3,475 to nearest hundred: look at tens digit ($7 \geq 5$) → round up → **3,500** (1 pt)
- 8.** School B (4,253) has more students. Compare tens: $5 > 3$. Full credit requires showing place-by-place comparison and 1-sentence explanation. (1 pt)
- 9.** *Error analysis (2 pts).* The student is **wrong**. 4,367 rounded to nearest thousand → look at hundreds digit: $3 < 5$ → round *down* → correct answer is **4,000**. The student likely rounded up incorrectly. Full credit: correct answer + identifies the error (looked at wrong digit, or rounded wrong direction). Partial credit (1 pt): correct answer without explanation.
- 10.** *Multi-part reasoning (2 pts).* (a) Nearest ten: **3,850** (look at ones: $7 \geq 5$, round up). (b) Nearest hundred: **3,800** (look at tens: $4 < 5$, round down). (c) Nearest thousand: **4,000** (look at hundreds: $8 \geq 5$, round up). Explanation: the answers differ because each rounding looks at a different digit in a different place. Full credit: all three correct + explanation mentioning different place values. Partial (1 pt): 2 of 3 correct or explanation only.

- ☐ **Units 3.02–3.12:** Answer guides for remaining units should be added here as each unit's CC and SA are finalized. Follow the same format: list each problem with the correct answer, key concept to look for in shown work, and partial-credit criteria for explanation items. [VERIFY — add answer keys here as units are completed and assessments are reviewed.]

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Teacher Workflow & Workload Map

Weekly routine · heavy units · announcement templates

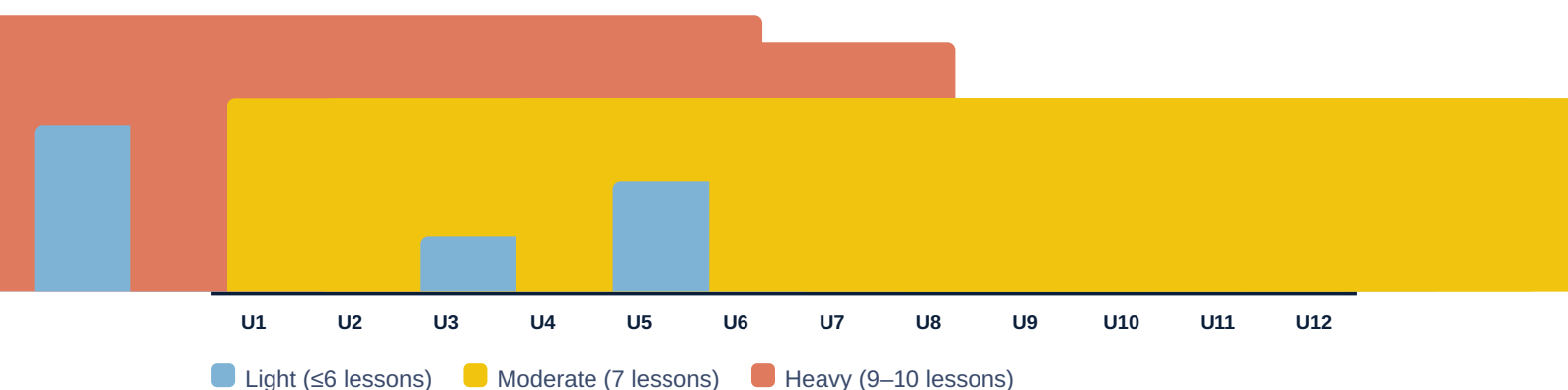
Normal-Week Teacher Tasks

Task	When	Time estimate
		~3–5 min/student

Review IP submissions in Canvas; spot-check 3–4 key problems per submission	After IP due date	
Grade Concept Checks (8 problems, use answer key in Section 6)	At unit midpoint	~5–8 min/student
Grade Summative Assessments (10 problems, 12 pts)	At unit end	~8–12 min/student
Review Spotlight Task uploads (Weeks 17–32)	Weekly	~2–3 min/student
Post weekly Canvas announcement	Monday morning	~10 min (use templates below)
Respond to student/family messages	Ongoing	Variable

Workload by Unit — Relative Teacher Grading Load

Units with more lessons and a full SA are heavier grading weeks. Units 3.02 and 3.04 (10 and 9 lessons respectively) are the longest. Units 3.07 (2 lessons) is the lightest.



Weekly Announcement Templates

Copy, customize the bracketed fields, and post in Canvas each Monday. Keep announcements short — students are 3rd graders working independently.

Template: Normal Week

Subject: Week [#] — What we're working on this week □

Hi 3rd Grade Math Explorers! □

This week we're working on **[lesson title(s) — e.g., "mental addition strategies"]**.
Open your Canvas Math module and work through the lesson page(s) before
completing your Independent Practice.

Don't forget: your Math Journal is your thinking space. Use it during the Warm-Up and
Learn tabs — even if nobody collects it, it helps your brain.

Due this week: [IP PDF for Lesson X.XX.XX — due [date]]

If you get stuck, send me a message and we'll figure it out together.

— [Your name]

Template: Concept Check Week

Subject: Week [#] — Concept Check coming up □

Hi explorers! This week you'll take the **Unit [X] Concept Check**. It covers [brief list —
e.g., "place value, expanded form, comparing numbers, and the number line"] —
everything from the first part of Unit [X].

The Concept Check is printed (or uploaded as a PDF). Show your work in the work box
for each problem. I'm looking at your thinking, not just your answers.

Before you take it: Flip back through your Math Journal. If expanded form or
comparing felt tricky, try one more practice problem on the lesson page first.

Due: Concept Check PDF uploaded to Canvas by [date].

— [Your name]

Template: Summative Assessment Week

Subject: Week [#] — Unit [X] Summative Assessment □

You made it to the end of Unit [X]! This week's Summative Assessment covers everything from the unit. It's a little longer than the Concept Check — 10 problems, and two of them ask you to explain your thinking in writing.

For the "Find the Mistake" problem: read the other student's work carefully, decide if they're right or wrong, and explain what went wrong and what the correct answer is.

Take your time. Show every step. Remember Cora — check your work before you're done.

Due: SA PDF uploaded by [date].

— [Your name]

Template: Spotlight Task Week (Weeks 17–32)

Subject: Week [17–32] — Spotlight Task + VR this week □□

This week we have a Spotlight Task! Open the Spotlight Task page in Canvas and follow the directions inside. You'll do a short VR activity and then upload a file from your work.

What to upload: [Specific file type — e.g., "a screenshot of your VR scene" — add once VR content is finalized]

Due: Spotlight Task upload by [date].

Keep exploring — math is everywhere in VR! □

— [Your name]

Template: Re-engagement Nudge (for quiet students)

Subject: Checking in on you ☐ *(send as individual message, not announcement)*

Hi [Student name]! I noticed you haven't submitted [assignment name] yet. I wanted to check in — is everything okay? Sometimes a lesson feels confusing, and that's completely normal.

If you got stuck somewhere, let me know which tab or which problem felt hard and we'll work through it together. No judgment — just want to make sure you get the math.

— [Your name]

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Florida B.E.S.T. Standards Coverage Map

Teacher copy only — standards codes do not appear on student pages

Florida B.E.S.T. standard codes are visible in the lesson HTML info strip but are not the focus of student-facing language. The table below maps each primary standard to the units and lessons that address it. [VERIFY codes against the full Florida B.E.S.T. 3rd grade document — codes below are drawn from the lesson HTML files and are correct as built.]

MA.3.NSO.1.1

Read and write numbers to 10,000 in standard, expanded, and word form

U1 L1–L2

MA.3.NSO.1.2

Compare and order numbers to 10,000 using $>$, $<$, $=$; use a number line

U1 L3–L4

MA.3.NSO.1.3

Identify and complete number patterns; round to nearest 10, 100, 1000

U1 L5–L8

MA.3.NSO.2.1

Add and subtract multi-digit numbers using mental strategies and standard algorithm

U2 L1–L7, U3 L1–L3

MA.3.NSO.2.2

Estimate sums and differences

U3 L4–L5

MA.3.NSO.2.3

Multiply one-digit numbers by multiples of 10 and 100

U5 L1

MA.3.AR.1.1	Apply properties of multiplication; recall multiplication and division facts 0–10	U4 L1–L6, U8 L1–L7
MA.3.AR.1.2	Solve one- and two-step word problems involving multiplication and division	U4 L7–L9, U5 L6–L7, U6 L6–L7
MA.3.AR.2.1	Multiply two-digit by one-digit numbers with and without regrouping	U5 L2–L7
MA.3.AR.2.2	Divide two- and three-digit numbers by a one-digit divisor	U6 L1–L7
MA.3.FR.1.1	Identify unit fractions; represent fractions on a number line	U9 L1–L2
MA.3.FR.1.2	Compare fractions with like numerators or like denominators	U9 L3–L4
MA.3.FR.2.1	Generate equivalent fractions; simplify fractions	U10 L1–L3
MA.3.FR.2.2	Compare and order fractions; add and subtract fractions with like denominators	U10 L4–L7
MA.3.M.1.1	Measure length in metric units (m, cm, km); solve word problems	U11 L1–L4, L7
MA.3.M.1.2	Measure capacity (L, mL) and mass (kg, g)	U11 L5–L6
MA.3.DP.1.1	Collect and represent data using picture graphs, bar graphs, and tables	U7 L1–L2
MA.3.GR.1.1	Identify and classify polygons; properties of triangles and quadrilaterals	U12 L4–L6
MA.3.GR.1.2	Identify circles, angles (right angles), and use geometric tools (compass)	U12 L1–L3, L7
MTR.2.1	Demonstrate understanding using models and representations (throughout)	All units
MTR.5.1	Use patterns, structures, and mathematical relationships to solve problems	All units

Classical Moves That Are Actually in the Lessons

These are not abstract principles — they are features you can point to in every lesson HTML file.

Classical Move	Where It Appears	What It Does
Notice/Wonder before instruction	Warm-Up tab → Think About It section	Students observe a number, object, or situation and record what they notice and wonder <i>before</i> the concept is taught. This is narration before imitation — the classical sequence of observation preceding formal instruction.
CPA sequence (imitation before independence)	Learn tab → Touch It / See It / Write It panels	Every concept is taught Concrete first (build with discs), then Pictorial (read a chart), then Abstract (write the equation). Students imitate the concrete model before working independently in abstract form. This mirrors classical grammar-stage imitation.
Socratic questioning	Learn tab → Socratic Question box (labeled "Stop and think")	Each lesson has a genuine question that doesn't have an obvious answer — "Which digit has the greatest value, and how do you know?" Students write their thinking before the explanation is revealed. This is interrogative, not declarative, teaching.
Multiple perspectives on one problem	Math Crew thought bubbles in Worked Examples	The same problem is solved four different ways (concrete, pictorial, shortcut, verification) in every worked example. Classical education prizes understanding a thing from multiple angles over mastering a single algorithm.
Virtue thread — mathematical character	Math Crew characters as recurring types	Cora (the checker) models intellectual humility and precision. Felix models efficiency. Atlas models patience with concrete work. Hazel models the value of representing ideas visually. Together they model what a virtuous mathematical thinker

		looks like — careful, curious, creative, self-correcting.
Quote strips — classical opening	Below the header, above the info strip	Each lesson opens with a short mathematical truth as a quote ("A digit tells more than its name. Its place tells its value."). This is the classical practice of beginning a lesson with a statement worth examining — the opening of the Great Conversation, even at 3rd grade.

How to Talk About This With Families

If families ask why the lessons start with questions rather than explanations, or why students are asked to guess before they're taught: this is deliberate. Classical math education holds that a student who has wrestled with a problem — even incorrectly — understands the answer more deeply when it arrives than a student who was told the rule first. The Notice/Wonder prompt and the Readiness Check are not filler. They are the pedagogical setup for genuine understanding.

How to Cultivate Mathematical Culture Asynchronously

In an async course with young students, culture comes through consistency and warmth, not live discussion. A few practices that work:

- Reference the Math Crew by name in your Canvas announcements and feedback. "I saw some real Cora thinking in your work this week — you checked your rounding carefully."
- When returning graded work, name the thinking that worked, not just the answer that was right. "You compared place by place starting from the thousands — that's exactly the strategy."
- Occasionally share a quote from a student's explanation (anonymized and with permission) in your weekly announcement to model good mathematical writing for the cohort.
- Hold the question open. When a student gets something wrong, respond with a question: "Which digit did you look at when you rounded? Let's start there."

